

Critical Thinking Assignment 1: Create MySQL Database

Alexander Ricciardi

Colorado State University Global

ITS410: Database Management

Dr. Murthy Rallapalli

April 20, 2025

Critical Thinking Assignment 1: Create MySQL Database

This documentation is part of the Critical Thinking 1 Assignment from ITS410: Database Management at Colorado State University Global. The documentation provides screenshots showcasing the creation of the My Guitar Shop database using a script and additional scripts illustrating basic data querying.

The Assignment Direction:

Create MySQL Database

You will use MySQL Workbench to create the My Guitar Shop database to review the tables in this database and to enter SQL statements and run them against this database.

Make sure the MySQL server is running

1. Start MySQL Workbench and open a connection for the root user.
2. Check whether the MySQL server is running. If it isn't, start it.

Use MySQL Workbench to create the My Guitar Shop database.

3. Download and open the script file named **my_guitar_shop.sql** by clicking the Open SQL Script File button in the SQL Editor toolbar. Then, use the resulting dialog box to locate and open the file.
4. Execute the entire script by clicking the Execute SQL Script button in the SQL editor toolbar or by pressing Ctrl+Shift+Enter. When you do, the Output window displays messages that indicate whether the script executed successfully. Take a screenshot.

Use MySQL Workbench to enter and run SQL statements

5. Double-click on the my_guitar_shop database to set it as the default database. When you do that, MySQL Workbench should display the database in bold.
6. Open a SQL editor tab. Then, enter and run this SQL statement:

SELECT product_name FROM products

Take a resulting screenshot.

7. Delete the *e* at the end of product_name and run the statement again. Note the error number and the description of the error. Take a resulting screenshot.
8. Open another SQL editor tab. Then, enter and run this statement:

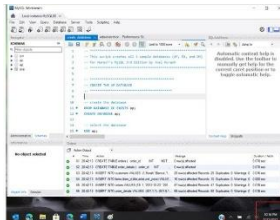
SELECT COUNT(*) AS number_of_products

FROM products

Use MySQL Workbench to open and run scripts

9. Download and open the script named **product_details.sql**. Note that this script contains just one SQL statement. Then, run the statement. Take a resulting screenshot.
10. Download and open the script named **product_summary.sql**. Note that this opens another SQL editor tab. Then, run the statement. Take a resulting screenshot.
11. Download and open the script named **product_statements.sql**. Notice that this script contains two SQL statements that end with semicolons. Then, run the statement. Take a resulting screenshot.

All the screenshots should show current date. Example of screenshot.

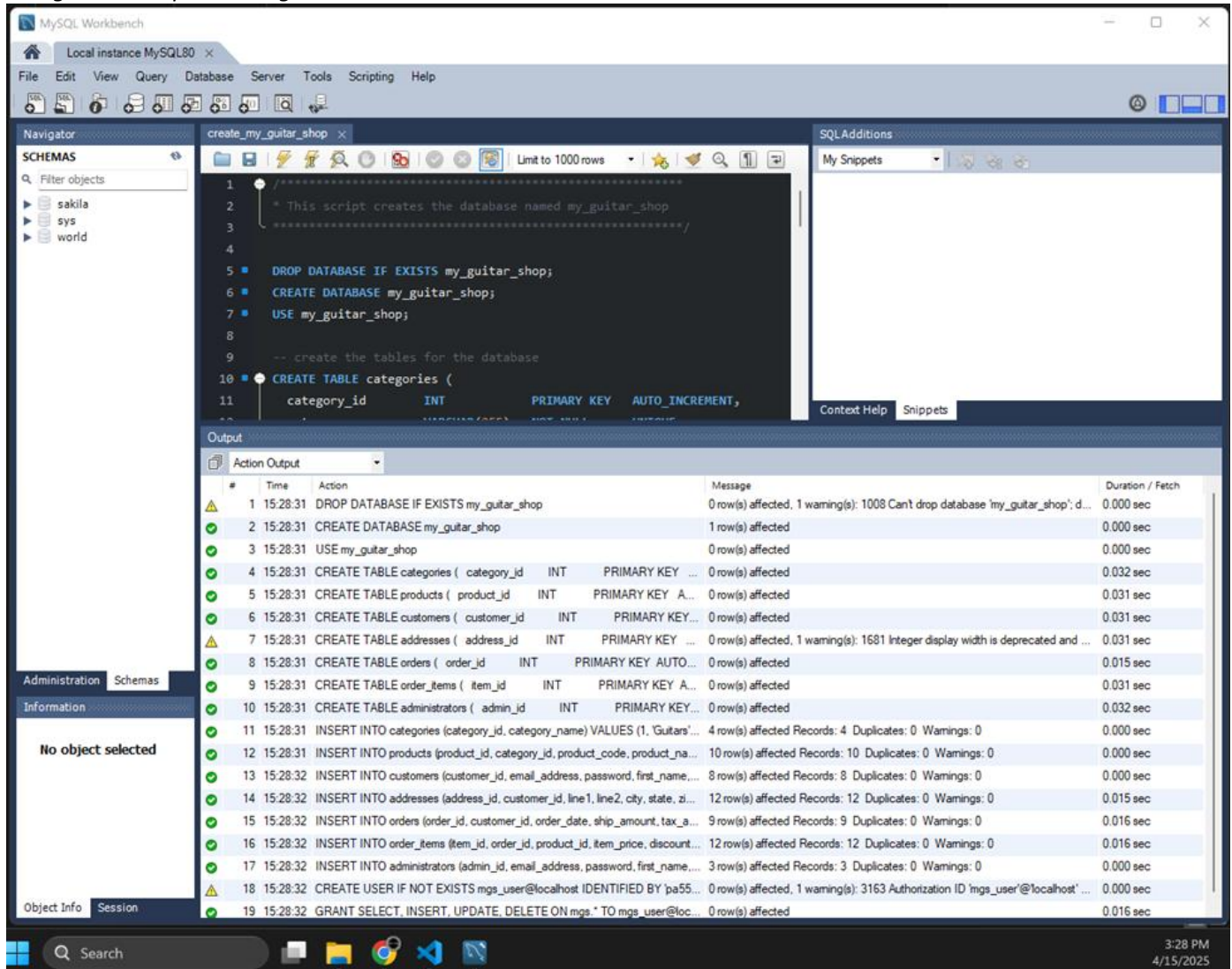


Submit your labeled results screenshots in a Word file.

Screenshots

Figure 1

Assignment Steps 1 through 4: Table Creation



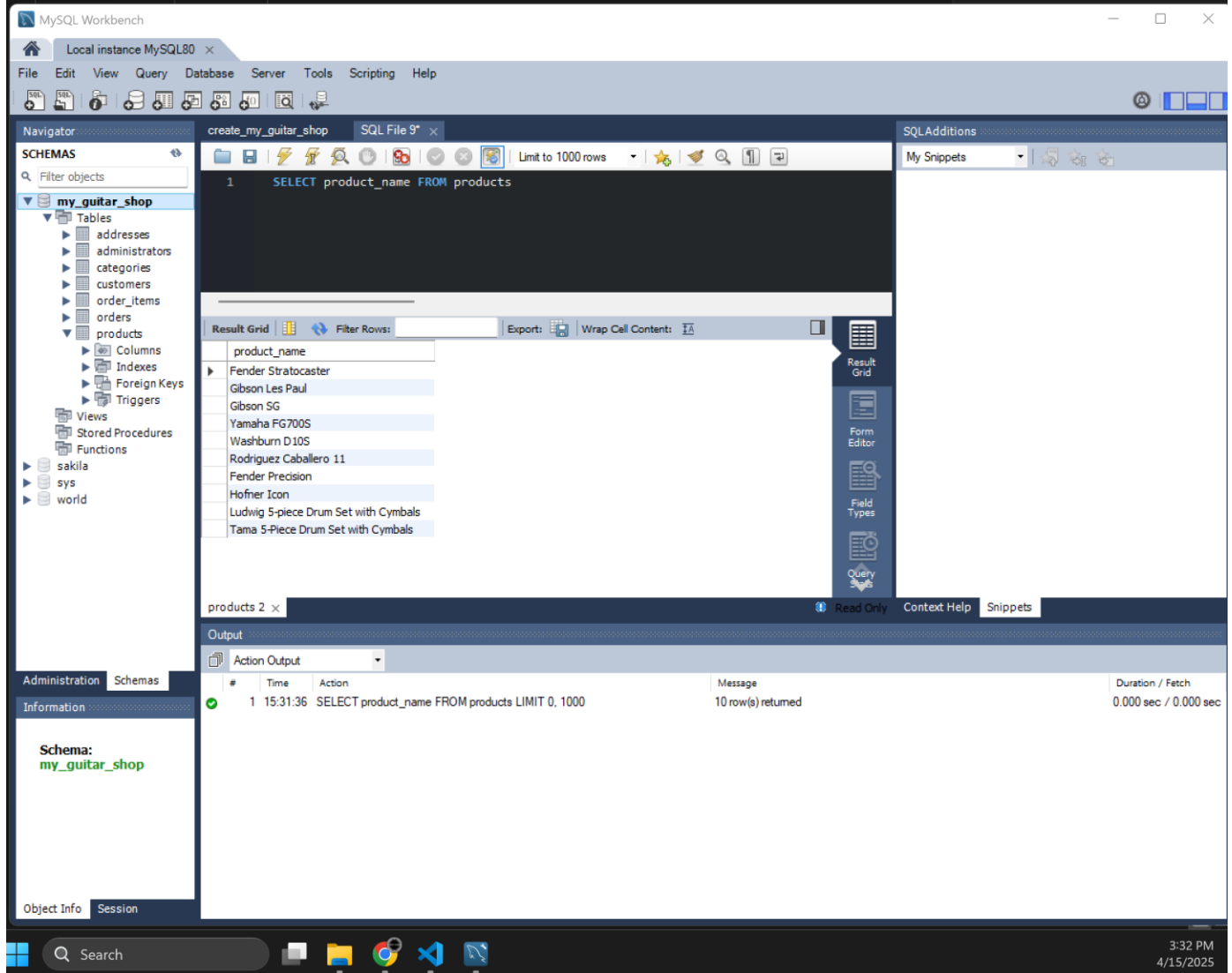
The screenshot displays the MySQL Workbench interface. The main editor window shows a SQL script named 'create_my_guitar_shop.sql'. The script includes comments and SQL commands to create a database named 'my_guitar_shop' and several tables: 'categories', 'products', 'customers', 'addresses', 'orders', 'order_items', and 'administrators'. The script also includes data insertion statements for each table.

The Output window at the bottom shows the results of the script execution. It lists the actions performed, the time taken, and the message returned by the database. The output shows that the database 'my_guitar_shop' was created successfully, and all tables were created without errors. Data was also inserted into each table successfully.

#	Time	Action	Message	Duration / Fetch
1	15:28:31	DROP DATABASE IF EXISTS my_guitar_shop	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'my_guitar_shop'; d...	0.000 sec
2	15:28:31	CREATE DATABASE my_guitar_shop	1 row(s) affected	0.000 sec
3	15:28:31	USE my_guitar_shop	0 row(s) affected	0.000 sec
4	15:28:31	CREATE TABLE categories (category_id INT PRIMARY KEY AUTO_INCREMENT,	0 row(s) affected	0.032 sec
5	15:28:31	CREATE TABLE products (product_id INT PRIMARY KEY A...	0 row(s) affected	0.031 sec
6	15:28:31	CREATE TABLE customers (customer_id INT PRIMARY KEY...	0 row(s) affected	0.031 sec
7	15:28:31	CREATE TABLE addresses (address_id INT PRIMARY KEY ...	0 row(s) affected, 1 warning(s): 1681 Integer display width is deprecated and ...	0.031 sec
8	15:28:31	CREATE TABLE orders (order_id INT PRIMARY KEY AUTO...	0 row(s) affected	0.015 sec
9	15:28:31	CREATE TABLE order_items (item_id INT PRIMARY KEY A...	0 row(s) affected	0.031 sec
10	15:28:31	CREATE TABLE administrators (admin_id INT PRIMARY KEY...	0 row(s) affected	0.032 sec
11	15:28:31	INSERT INTO categories (category_id, category_name) VALUES (1, 'Guitars')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
12	15:28:31	INSERT INTO products (product_id, category_id, product_code, product_na...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
13	15:28:32	INSERT INTO customers (customer_id, email_address, password, first_name,...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.000 sec
14	15:28:32	INSERT INTO addresses (address_id, customer_id, line1, line2, city, state, zi...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0	0.015 sec
15	15:28:32	INSERT INTO orders (order_id, customer_id, order_date, ship_amount, tax_a...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0.016 sec
16	15:28:32	INSERT INTO order_items (item_id, order_id, product_id, item_price, discount...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0	0.016 sec
17	15:28:32	INSERT INTO administrators (admin_id, email_address, password, first_name...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
18	15:28:32	CREATE USER IF NOT EXISTS mgs_user@localhost IDENTIFIED BY 'pa55...	0 row(s) affected, 1 warning(s): 3163 Authorization ID 'mgs_user'@'localhost' ...	0.000 sec
19	15:28:32	GRANT SELECT, INSERT, UPDATE, DELETE ON mgs.* TO mgs_user@loc...	0 row(s) affected	0.016 sec

Note: The figure illustrates the MySQL Workbench result after performing steps 1 through 4, my_guitar_shop table creation. Line 19 is the last line in the Output window, showing a log of the actions performed when executing the MySQL script (create_my_guitar_shop.sql)

Please see the next page.

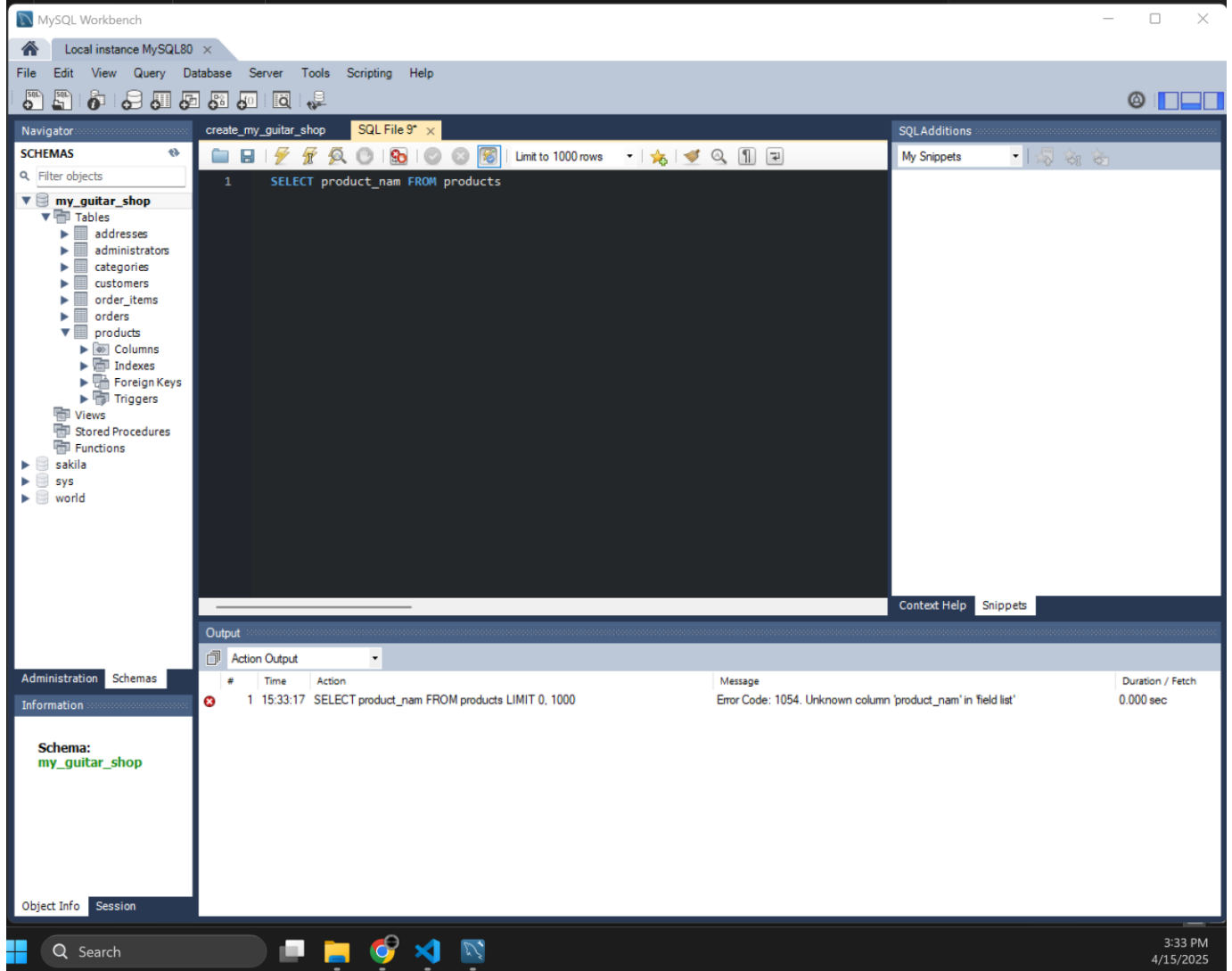
Figure 2*Assignment Steps 5 through 6: Product Names*

Note: The figure illustrates the MySQL Workbench result after performing steps 5 through 6: Product Names.

Please see the next page.

Figure 3

Assignment Step 7: Product Names Error (Screenshot Not Required by Assignment)

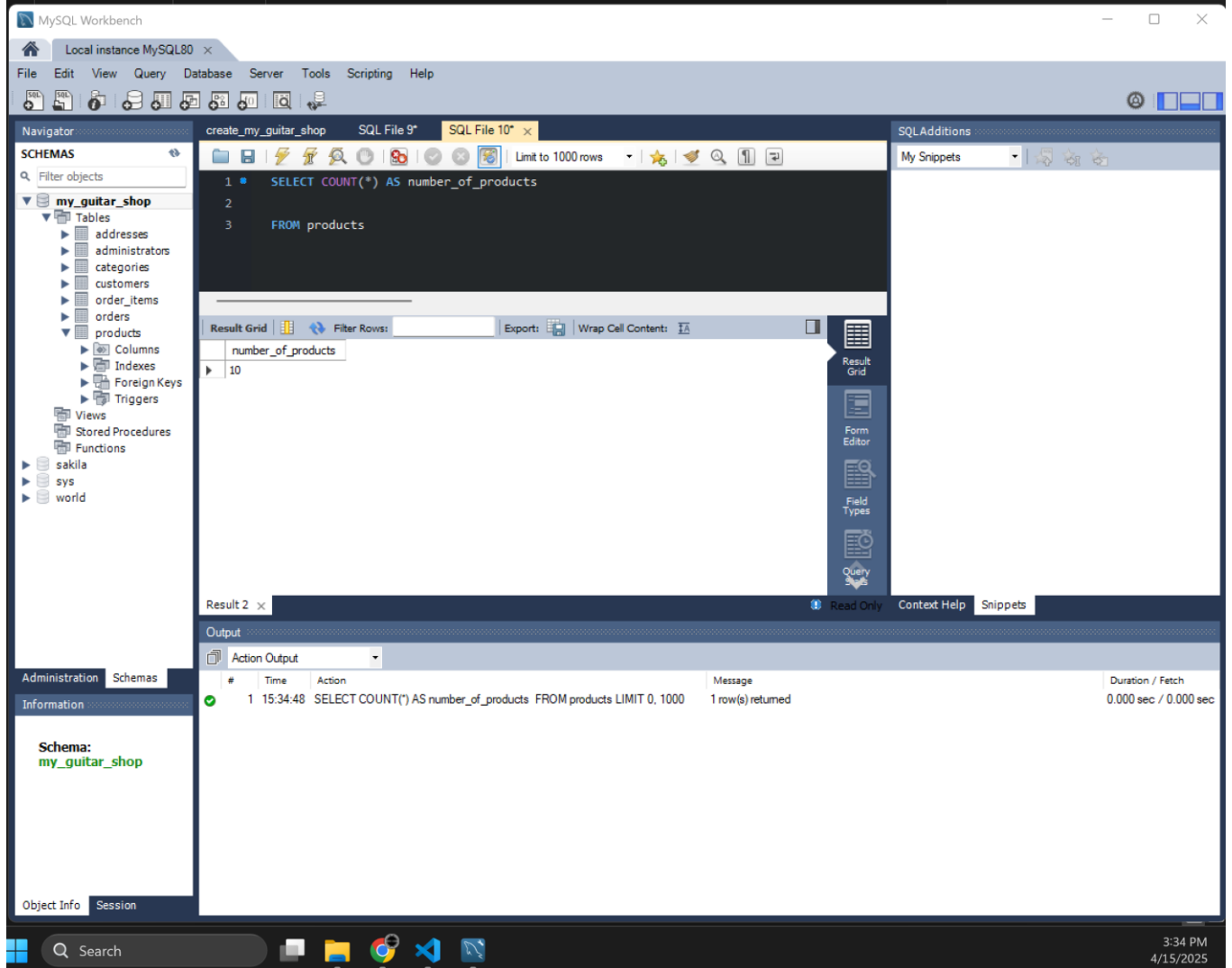


Note: The figure illustrates the MySQL Workbench result after performing steps 7: Product Names Error (Screenshot Not Required by assignment).

Please see the next page.

Figure 4

Assignment Step 8: Number of Products (Screenshot Not Required by Assignment)



Note: The figure illustrates the MySQL Workbench result after performing step 8: Number of Products (Screenshot Not Required by assignment).

Please see the next page.

Figure 5
Assignment Step 9: Product details

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'my_guitar_shop' selected. The main editor shows a SQL query: `SELECT product_name, list_price, date_added FROM products ORDER BY product_name`. The 'Result Grid' displays the following data:

product_name	list_price	date_added
Fender Precision	799.99	2018-06-01 11:29:35
Fender Stratocaster	699.00	2017-10-30 09:32:40
Gibson Les Paul	1199.00	2017-12-05 16:33:13
Gibson SG	2517.00	2018-02-04 11:04:31
Hofner Icon	499.99	2018-07-30 14:18:33
Ludwig 5-piece Drum Set with Cymbals	699.99	2018-07-30 12:46:40
Rodriguez Caballero 11	415.00	2018-07-30 14:12:41
Tama 5-Piece Drum Set with Cymbals	799.99	2018-07-30 13:14:15
Washburn D10S	299.00	2018-07-30 13:58:35
Yamaha FG700S	489.99	2018-06-01 11:12:59

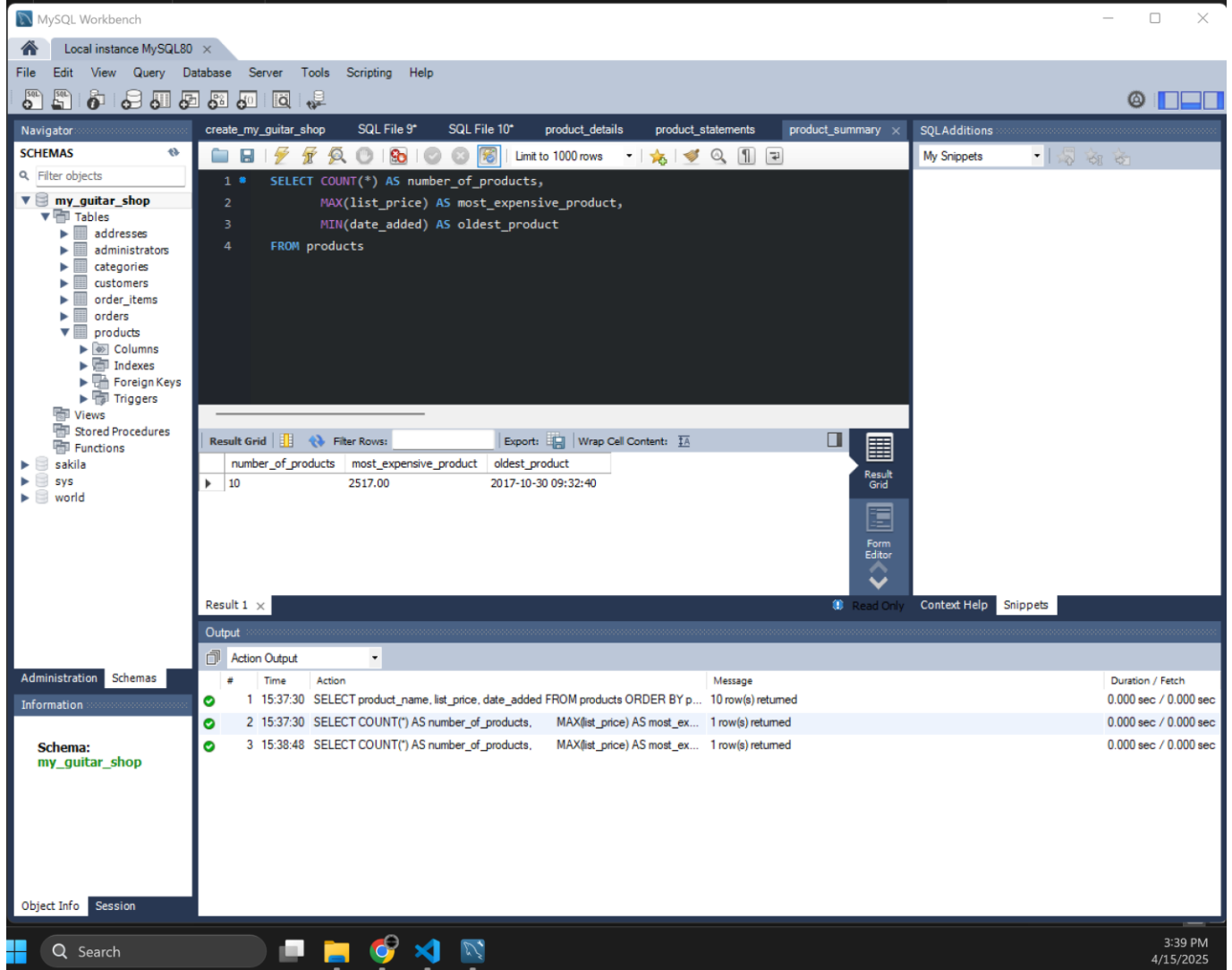
The bottom panel shows the 'Output' tab with the following action output:

#	Time	Action	Message	Duration / Fetch
1	15:34:48	SELECT COUNT(*) AS number_of_products FROM products LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
2	15:35:46	SELECT product_name, list_price, date_added FROM products ORDER BY p...	10 row(s) returned	0.000 sec / 0.000 sec

Note: The figure illustrates the MySQL Workbench result after performing step 9: Product details.

Please see the next page.

Figure 6
Assignment Step 10: Product Summary



Note: The figure illustrates the MySQL Workbench result after performing step 10: Product Summary.

Please see the next page.

Figure 7
Assignment Step 11: Product Statement

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'my_guitar_shop' schema with various tables. The main editor window shows the following SQL queries:

```

1 SELECT product_name, list_price, date_added
2 FROM products
3 ORDER BY product_name;
4
5 SELECT COUNT(*) AS number_of_products,
6        MAX(list_price) AS most_expensive_product,
7        MIN(date_added) AS oldest_product
8 FROM products;

```

The 'Result Grid' shows the output of the second query:

number_of_products	most_expensive_product	oldest_product
10	2517.00	2017-10-30 09:32:40

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	15:37:30	SELECT product_name, list_price, date_added FROM products ORDER BY p...	10 row(s) returned	0.000 sec / 0.000 sec
2	15:37:30	SELECT COUNT(*) AS number_of_products, MAX(list_price) AS most_ex...	1 row(s) returned	0.000 sec / 0.000 sec

Note: The figure illustrates the MySQL Workbench result after performing step 11: Product Statement.